Week 6 — Probabilistic Models & Bayesian Inference

Resource Guide & Learning Toolkit

Statistical Machine Learning · Fusemachines AI Fellowship

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How to use this guide
This document is organised in six sections that mirror the six learning phases of the module. Work through each phase in order. **Step 1 — Understand the topic** by watching the linked YouTube videos (they are short and concept-focused). **Step 2 — Go deeper** with the textbook chapters or documentation pages listed. **Step 3 — Practise with AI prompts**: paste each prompt into ChatGPT or Claude, then use the critique guidance to evaluate and challenge the response. **Step 4 — Discuss**: bring the discussion questions to your mid-week sync. All links in this document are clickable — click any URL to open it directly. Resources marked ★ **Priority** are the minimum required before the session.

LEARNING PHASES

Core topics and subtopics across six progressive phases

Phase	Topics & subtopics
1 · The Estimation Trinity: MLE, MAP & Full Bayes	• Maximum Likelihood Estimation (MLE) • Maximum A Posteriori (MAP) estimation • Full Bayesian posterior inference • Prior × likelihood → posterior; the normalising constant • Beta–Binomial conjugacy • Credible intervals (HDI) vs. frequentist confidence intervals • Monte Carlo estimation of a comparison probability, e.g. $P(\theta_A > \theta_B)$
2 · Sequential Updating & Dirichlet–Multinomial	• Sequential (online) Bayesian updating — the posterior becomes the next prior • The Beta–Binomial update rule • Posterior evolution as n grows; prior wash-out • Decision thresholds read from the posterior, e.g. $P(\theta > c)$ • Dirichlet–Multinomial conjugacy (multi-category extension) • Concentration parameters and pseudo-counts
3 · Multivariate Gaussians	• The multivariate normal: mean vector and covariance matrix • Covariance vs. correlation; confidence ellipses (1σ / 2σ / 3σ) • Conditional Gaussian distributions • Marginalisation of a joint Gaussian • Condition number $\kappa(\Sigma)$ and its link to multicollinearity
4 · Probabilistic Graphical Models	• Directed graphical models: Bayesian Networks (DAGs) • Conditional independence and d-separation • Exact inference: variable elimination and belief propagation • Forward (predictive) vs. backward (diagnostic) inference • Undirected models: Markov Random Fields and pairwise factors • Directed vs. undirected — implications for causal interpretation
5 · Gaussian Process Regression	• Gaussian Processes as priors over functions • Kernel (covariance) functions: RBF, periodic (ExpSineSquared), white-noise, linear (DotProduct) • Composite (additive) kernels: trend + seasonality + noise • Posterior predictive mean and credible bands • Interpolation vs. extrapolation; how uncertainty grows away from data • Hyperparameter learning via the marginal likelihood
6 · MCMC: Bayesian Logistic Regression	• Sampling from intractable posteriors with MCMC • Hamiltonian Monte Carlo and the NUTS sampler (PyMC) • Weakly-informative priors for regression coefficients • Feature scaling and posterior geometry • Convergence diagnostics: R-hat, effective sample size (ESS), divergences • Prior sensitivity analysis • Bayesian HDI vs. frequentist CI; saving / serialising the trace

RESOURCES

Curated materials — click any link to open directly

Reading strategy: If you have 1–2 hours, watch the priority videos first — they give you the mental model. If you have 4+ hours, pair each video with its corresponding textbook chapter. For Phases 1–2, Bishop PRML Ch. 2 is the definitive reference but is dense — read it twice. For Phase 6, Bayesian Methods for Hackers Ch. 1–3 builds the intuition you need before touching PyMC. The Rasmussen & Williams GP book Chapter 2 covers everything needed for Phase 5 in one self-contained chapter.

★ Priority Videos — Watch Before the Session

➤ Bayes theorem, the geometry of changing beliefs — 3Blue1Brown

https://youtu.be/HZGCoVF3YvM

★ Priority · Phases 1–2. The best visual intuition for Bayesian updating — how the posterior is the normalised product of prior and likelihood. Essential before any formula work. **Watch this first.**

➤ MLE and MAP — StatQuest

https://youtu.be/Dn6b9fCIUpM

★ Priority · Phase 1. Explains maximum likelihood and MAP estimation clearly with examples. Bridges the gap between frequentist and Bayesian estimation.

➤ Binomial distributions — 3Blue1Brown (Probabilities of probabilities, part 1)

https://youtu.be/8idr1WZ1A7Q

★ Priority · Phase 2. Visual prerequisite for the Beta-Binomial conjugacy section — builds intuition for the binomial and for reasoning about a distribution over a probability. Watch before deriving the posterior update rule.

➤ MCMC Explained — ritvikmath

https://youtu.be/yApmR-c_hKU

★ Priority · Phase 6. Builds intuition for Markov Chain Monte Carlo from scratch. Covers the Metropolis-Hastings algorithm and why chains need a warm-up period.

➤ Gaussian Processes — Nando de Freitas (Oxford lecture)

https://youtu.be/4vGiHC35j9s

★ Priority · Phase 5. The most pedagogically clear lecture on GPs — explains function-space priors, the kernel trick, and posterior predictive distributions. Watch before coding.

➤ Bayesian Networks — CMU 10-708 PGM

https://youtu.be/TuGDMj43ehw

★ Priority · Phase 4. Full lecture covering DAGs, conditional independence, d-separation, and variable elimination. Essential preparation for the pgmpy exercises.

➤ A Quickstart Guide to PyMC — PyData (Nicole Carlson)

https://youtu.be/rZvro4-nFIk

Phase 6. Conference talk introducing the PyMC modelling workflow: model specification, sampling, and diagnostics. (PyMC3-era API — pair with the official PyMC docs for current v5 syntax.)

Textbooks & Long-Form Reading

● Pattern Recognition and Machine Learning — Bishop (2006), Ch. 1–3, 8

https://www.microsoft.com/en-us/research/wp-content/uploads/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf

★ Priority if you have time · Phases 1–4. Free PDF. The foundational reference for all probabilistic ML. Study Ch. 2 (Probability Distributions) first — it covers MLE, MAP, Beta-Binomial, Dirichlet, and the Exponential Family. Ch. 8 covers PGMs.

● Bayesian Methods for Hackers — Cameron Davidson-Pilon

https://github.com/CamDavidsonPilon/Probabilistic-Programming-and-Bayesian-Methods-for-Hackers

★ Priority for Phase 6 · Free online. The best resource for building PyMC intuition quickly. Ch. 1–3 covers everything needed for the MCMC section with working code examples. Read before touching Phase 6.

● Gaussian Processes for Machine Learning — Rasmussen & Williams

https://gaussianprocess.org/gpml/

★ Priority for Phase 5 · Free PDF. The authoritative GP reference. Chapter 2 alone covers GP regression, kernel functions, and posterior predictive distributions — everything needed for the assignment. Self-contained.

● Machine Learning: A Probabilistic Perspective — Murphy (2012), Ch. 3, 5, 7

https://probml.github.io/pml-book/book0.html

Phases 1–3. Free online. Ch. 3 covers generative models for discrete data. Ch. 5.3 covers Bayesian logistic regression. Complements Bishop with more modern notation.

● Think Bayes — Allen Downey (free online)

https://allendowney.github.io/ThinkBayes2/

Phases 1–2. Free online. Code-first introduction to Bayesian reasoning. Excellent for students who find Bishop's mathematical density difficult — builds the same concepts with runnable Python examples.

Official Documentation & Reference

§ PyMC Documentation & Tutorial

https://www.pymc.io/welcome.html

Phase 6 — Essential. Official API reference for model specification, sampling, and the full PyMC v5 ecosystem. Read the Getting Started guide before the session.

§ ArviZ: Exploratory Analysis of Bayesian Models

https://python.arviz.org/en/stable/

Phase 6. API reference for az.plot_trace(), az.summary(), az.hdi(). Use this when generating convergence diagnostics.

§ pgmpy: Python library for PGMs

https://pgmpy.org/

Phase 4. API reference for DiscreteBayesianNetwork, DiscreteMLE, VariableElimination, DiscreteMarkovNetwork, and BeliefPropagation.

§ scikit-learn: Gaussian Process Regression

https://scikit-learn.org/stable/modules/gaussian_process.html

Phase 5. Official user guide for GaussianProcessRegressor, RBF, ExpSineSquared, and WhiteKernel. Includes the Mauna Loa example.

§ statsmodels — Mauna Loa CO_2 Dataset

https://www.statsmodels.org/stable/datasets/generated/co2.html

Phase 5. Dataset documentation — 521 months of atmospheric CO_2 (1958–2001). Load in 3 lines: from statsmodels.datasets import co2; df = co2.load_pandas().data.

§ PyMC Discourse Community

https://discourse.pymc.io/

Phase 6. The authoritative community for debugging MCMC issues. Search for divergences, R-hat > 1.01, and ESS problems before asking new questions.

Datasets

◆ Telco Customer Churn Dataset · Kaggle

https://www.kaggle.com/datasets/blastchar/telco-customer-churn/data

7,043 rows · 21 columns · same CSV from Week 5. Used for Parts 1, 2, 3, 4, and 6. No new EDA required — you already know this data. Known issues: TotalCharges whitespace nulls (coerce to numeric), ~27% churn rate (class imbalance).

◆ Mauna Loa Atmospheric CO_2 — statsmodels built-in

https://www.statsmodels.org/stable/datasets/generated/co2.html

521 months (1958–2001) · Part 5 only · zero download required. Load in 3 lines: from statsmodels.datasets import co2. No CSV, no Kaggle, no preprocessing. The canonical GP regression dataset: slow upward trend + sharp annual cycle + measurement noise.

AI PROMPTS & CRITIQUE GUIDE

How to use a language model as a learning tool — not a shortcut

How to use these prompts: Paste each prompt into Claude or ChatGPT. Read the response carefully, then apply the **Critique / Verify** guidance below it to actively evaluate whether the answer is correct, incomplete, or misleading. The goal is **critical engagement** — you are the expert checking the AI's work, not a passive reader. If the model gets something wrong, ask a follow-up that forces it to fix the mistake. Write a short note in your notebook about what it got right and what it missed.

Phase 1 — MLE, MAP & Full Bayes

Ask: "Explain the difference between a frequentist confidence interval and a Bayesian credible interval — are they the same thing?"

Critique / Verify: The model must correctly explain that a frequentist 95% CI makes a statement about the **procedure** (95% of such intervals will contain the true parameter across repeated experiments), while a Bayesian 94% HDI makes a direct probability statement about the parameter **given the data observed**. Many LLMs conflate these. Does it give a concrete numerical example? Does it correctly state that you cannot say 'there is a 95% probability the parameter is in this interval' for a frequentist CI? Push back if it glosses over.

Ask: "Explain conjugate priors: why does a Beta prior with a Binomial likelihood give a Beta posterior?"

Critique / Verify: Check: does it derive the posterior algebraically by writing $P(\theta|k,n) \propto P(k|\theta,n) \cdot P(\theta)$ and completing the Beta kernel — or does it just state the result? Does it identify which terms are proportionality constants absorbed into Z? Does it mention the Exponential Family connection? Ask it to show the full algebraic derivation step by step.

Phase 4 — Probabilistic Graphical Models

Ask: "Write Python code using PyMC to perform Bayesian logistic regression on a binary classification problem."

Critique / Verify: Does the model specify weakly informative priors (e.g., Normal(0, 2)) or flat uninformative ones (pm.Flat())? Flat priors on logistic regression coefficients create improper posteriors and cause NUTS to diverge. Did the LLM warn you? Does it include feature scaling before passing data to the model? Does it use target_accept=0.9?

Ask: "Design a Gaussian Process kernel for a time series with a long-term trend, an annual seasonal cycle, and measurement noise."

Critique / Verify: Does it propose a composite kernel: RBF + RBF×ExpSineSquared + WhiteKernel? Does it justify each component? Does it recommend fixing the periodicity hyperparameter to 1 year rather than letting it optimise freely (which can be numerically unstable)? If the model proposes a single RBF kernel, it is underpowered — force it to explain why each component is necessary.

MID-WEEK DISCUSSION QUESTIONS

Designed to surface misconceptions — bring your reasoning, not just your answer

How to use these questions: These are deliberately open-ended and adversarial — they are designed so that a superficial answer reveals a gap. Before the discussion, write a 2–3 sentence answer in your notebook. During the session, compare your reasoning with peers. There is rarely a single right answer; the quality of your justification matters more than the conclusion.

Q1. Prior Dominance vs Data Dominance

Suppose a Beta–Binomial model with a weakly-informative Beta prior and a small sample (say n≈40) gives — for example — MAP≈0.39 and MLE≈0.43 (illustrative values). A colleague dismisses the prior as 'making up data.' How would you defend a weakly-informative prior using domain knowledge about churn rates? At what sample size does such a prior become essentially irrelevant, and how would you compute that analytically?

Q2. The Credible Interval vs Confidence Interval Battle

Suppose you present to the VP and say: 'there is a 94% probability the Month-to-month churn rate lies between, say, 0.38 and 0.49' (example numbers). A colleague replies: 'you cannot say that.' Who is technically correct? Under which statistical framework is each statement valid? Which is more useful for the VP's decision to launch a retention campaign?

Q3. GP Kernel Ablation

Suppose a GP for a trended time series uses an RBF kernel for the trend component. A colleague argues for a Matérn 3/2 kernel instead, because the signal is not infinitely smooth. What is the mathematical difference between the two kernels in terms of differentiability of sample functions? How would you decide between them empirically?

Q4. MCMC Pathology Diagnosis

Suppose you run 4 MCMC chains and — for example — three show R-hat≈1.00 with ESS in the high hundreds, while one shows R-hat≈1.38 with ESS≈55 (illustrative values). List the three most likely causes of this divergence. Which ArviZ diagnostic plots would you examine first?

Q5. ESS vs Visual Trace

Suppose a colleague runs MCMC with very few draws (say draws=200) and claims 'it converged — the trace looks like a fuzzy caterpillar', yet az.summary() reports a tiny effective sample size (e.g. ESS≈18) for the intercept. Why is such a low ESS a serious problem even when the trace looks visually acceptable? What is the relationship between MCMC autocorrelation and effective sample size?

Q6. BN Structure vs Data-Driven Learning

Suppose a Bayesian Network is hand-specified using domain knowledge, and a colleague argues for running a structure-learning algorithm (PC algorithm or score-based search) on the data instead. What are the risks given (a) class imbalance (e.g. a roughly 1:3 positive rate), (b) near-collinearity among related features such as tenure / MonthlyCharges / TotalCharges, and (c) the fact that correlation does not imply causal direction?

Q7. Small-Data Bayesian vs Frequentist

Suppose you fit both frequentist and Bayesian logistic regression on a small subset (say n≈250) and get — for example — a frequentist coefficient β≈0.08 (p≈0.12) and a Bayesian posterior mean ≈0.07 with a 94% HDI of roughly [–0.03, 0.17] (illustrative values). A colleague says 'the Bayesian model tells us nothing more.' What can you show from the full posterior that a p-value cannot express? Is there a business case to act on a coefficient whose HDI crosses zero?

Q8. GP Long-Horizon Forecasting Limits

Suppose a GP extrapolated well beyond its training window (say ~14 years) and the 95% credible band there is, for example, about ±11 ppm (illustrative). A policymaker asks: 'is that a lot of uncertainty?' Separate the uncertainty into epistemic and aleatoric components. Could you reduce the epistemic part with more training data, or is extrapolation uncertainty a structural property of the kernel's decay?

PHASE → RESOURCE MAP

At a glance: which resource to use for each phase

Phase	Topic	★ Watch	★ Read / Docs
1	MLE · MAP & Full Bayes	Bayes Theorem — 3B1B MLE and MAP — StatQuest	Bishop PRML Ch. 2 Think Bayes
2	Sequential Updating	Binomial distributions — 3B1B	Hackers Book Ch. 1–3
3	Multivariate Gaussians	3B1B: Essence of Linear Algebra	Bishop PRML Ch. 2.3

Phase	Topic	★ Watch	★ Read / Docs
4	Bayesian Networks & MRFs	CMU 10-708 PGM Lecture	pgmpy Docs Bishop Ch. 8
5	Gaussian Process Regression	GP Lecture — de Freitas	Rasmussen & Williams GP Book sklearn GP Docs
6	MCMC & Bayesian LR	MCMC — ritvikmath PyMC Quickstart — PyData	PyMC Docs ArviZ Docs